TEST INPUT - readiness trace pump panel

PP1

Job number	TEST-001
Issued under	iec-default
Design rules	software defaults, not confirmed by the company's engineers

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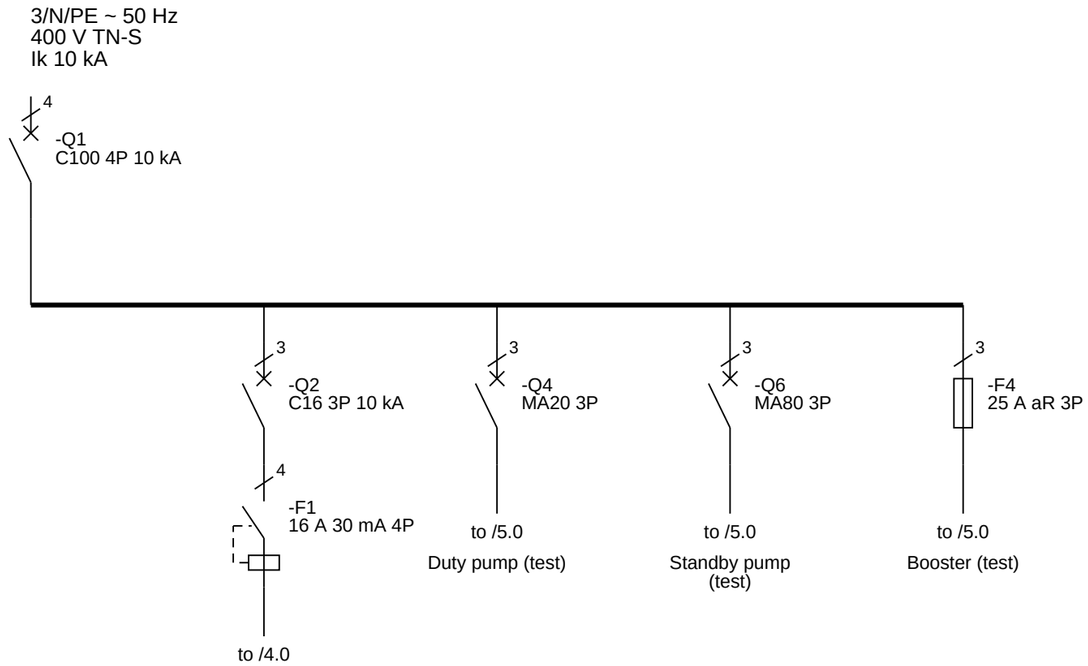

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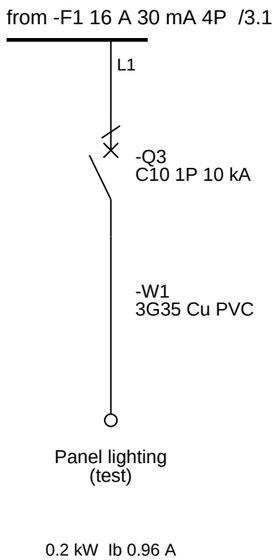

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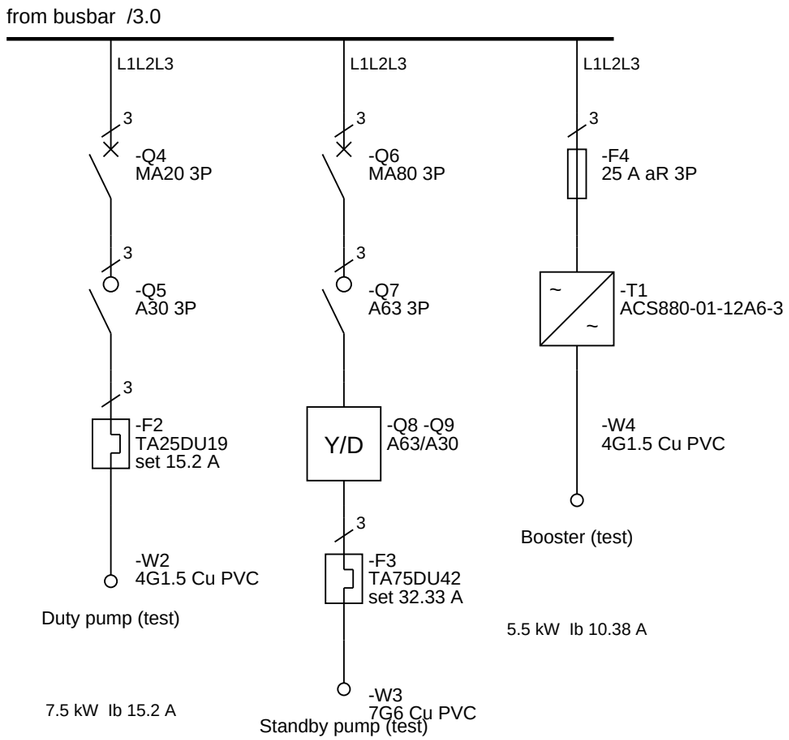

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No.	Note
1	Panel lighting (test): cable enlarged from 1.5 to 6 mm² for voltage drop over 900 m: 2.81 % from the origin, within the 3 % limit.
2	Panel lighting (test): cable enlarged from 6 to 35 mm² so a short circuit at its far end trips its breaker at once: Ikmin 133 A, at least 100 A (ABB handbook §2.4).
3	Group 1 breaker raised to 16 A, 1.6 x the 10 A breaker after it, for discrimination.
4	Duty pump (test): Ir 15.2 A, the typical Ir of a 7.5 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate.
5	Duty pump (test): dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2.
6	Standby pump (test): Ir 56 A, the typical Ir of a 30 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate.
7	Standby pump (test): star_delta starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 5: 400 V 50 kA Y/D Normal Type 2.
8	Standby pump (test): star and delta contactors need a mechanical interlock; the motor is fed by six conductors (U1 V1 W1, U2 V2 W2).
9	Booster (test): Ir 10.38 A from 5.5 kW shaft power at cos phi 0.85, efficiency 0.9; check it against the nameplate.
10	Booster (test): ACS880-01-12A6-3 for normal duty at 30 °C, behind 25 A aR fuses (ABB ACS880-01 hardware manual (3AUA0000078093)). Use a screened, symmetrical motor cable, earthed 360° at both ends.
11	Booster (test): the fuses need at least 75 A prospective short-circuit current at the board to operate fast enough.
12	Breakers are rated for at least 10 kA breaking capacity against 10 kA prospective at the board; each RCCB's conditional short-circuit current with the breaker ahead of it must reach it too.
13	Leave 1 spare outgoing ways (20 %).
14	Loads without a power factor were taken at cos phi 0.9.
15	Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.
16	No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze.
17	Each cable's withstand k²S² is in the cable list; compare it with the selected breaker's let-through I²t at 10 kA from its maker's curve (ABB handbook §2.4).
18	Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.
19	Circuit rules are this software's defaults, not confirmed by the company's engineers.
20	Enclosure: none named, since 11 devices have no width (rail_widths_mm in the company settings).

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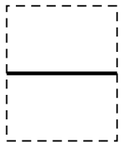
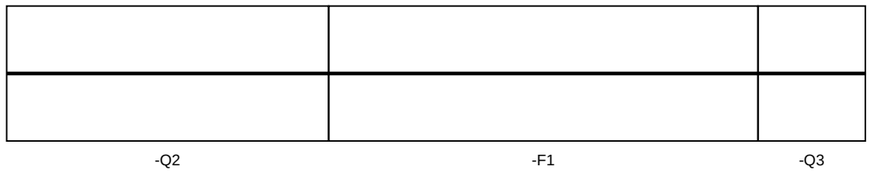
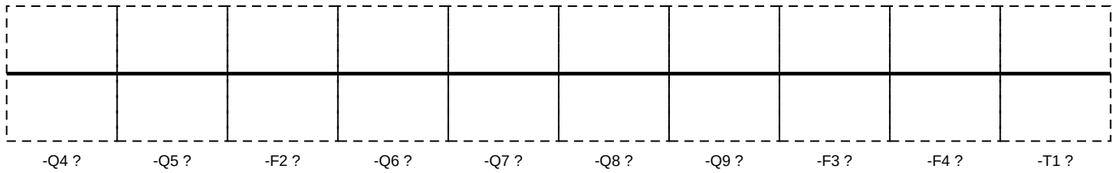
Incomer: 0 mm of rail; 1 without a width



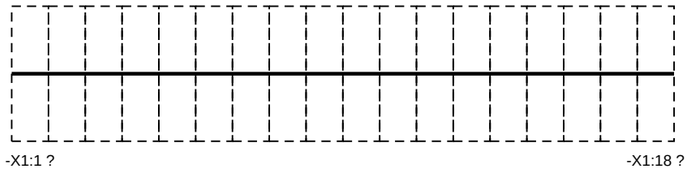
Residual current group: 140 mm of rail



Busbar: 0 mm of rail; 10 without a width



Terminals: 0 mm of rail; 18 without a width



Dashed: no width is given for the device's kind (rail_widths_mm in the company settings).

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Terminal	Conductor	Circuit	Cable	mm²	Article
-X1:1	L1	Panel lighting (test)	-W1	35	8WH1000-0AM00
-X1:2	N	Panel lighting (test)	-W1	35	8WH1000-0AM00
-X1:3	PE	Panel lighting (test)	-W1	35	8WH1000-0CM07
-X1:4	U	Duty pump (test)	-W2	1.5	8WH1000-0AF00
-X1:5	V	Duty pump (test)	-W2	1.5	8WH1000-0AF00
-X1:6	W	Duty pump (test)	-W2	1.5	8WH1000-0AF00
-X1:7	PE	Duty pump (test)	-W2	1.5	8WH1000-0CF07
-X1:8	U1	Standby pump (test)	-W3	6	8WH1000-0AK00
-X1:9	V1	Standby pump (test)	-W3	6	8WH1000-0AK00
-X1:10	W1	Standby pump (test)	-W3	6	8WH1000-0AK00
-X1:11	U2	Standby pump (test)	-W3	6	8WH1000-0AK00
-X1:12	V2	Standby pump (test)	-W3	6	8WH1000-0AK00
-X1:13	W2	Standby pump (test)	-W3	6	8WH1000-0AK00
-X1:14	PE	Standby pump (test)	-W3	6	8WH1000-0CK07
-X1:15	U	Booster (test)	-W4	1.5	8WH1000-0AF00
-X1:16	V	Booster (test)	-W4	1.5	8WH1000-0AF00
-X1:17	W	Booster (test)	-W4	1.5	8WH1000-0AF00
-X1:18	PE	Booster (test)	-W4	1.5	8WH1000-0CF07

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Cable	From	To	Type	Length
PP1 -W1	PP1 -Q3	Panel lighting (test)	3G35 Cu PVC	900 m
PP1 -W2	PP1 -Q4	Duty pump (test)	4G1.5 Cu PVC	30 m
PP1 -W3	PP1 -Q6	Standby pump (test)	7G6 Cu PVC	30 m
PP1 -W4	PP1 -F4	Booster (test)	4G1.5 Cu PVC	20 m

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Qty	Description	Article	Order number	Devices
1	Circuit-breaker C100 4P 10 kA	not selected		-Q1
1	Circuit-breaker C16 3P 10 kA	not selected		-Q2
1	Residual current circuit-breaker 16 A 30 mA 4P	not selected		-F1
1	Circuit-breaker C10 1P 10 kA	not selected		-Q3
1	Circuit-breaker MA20 3P	ABB T2S160 MA 20		-Q4
2	Contactor A30 3P	ABB A30		-Q5, -Q9
1	Overload relay TA25DU19 set 15.2 A	ABB TA25DU19		-F2
1	Circuit-breaker MA80 3P	ABB T2S160 MA80		-Q6
2	Contactor A63 3P	ABB A63		-Q7, -Q8
1	Overload relay TA75DU42 set 32.33 A	ABB TA75DU42		-F3
1	Fuse 25 A aR 3P	Bussmann 170M1311		-F4
1	Variable speed drive ACS880-01-12A6-3	ABB ACS880-01-12A6-3		-T1
2	Terminal 8WH1	Siemens 8WH1000-0AM00	8WH1000-0AM00	PP1 -X1
1	PE terminal 8WH1	Siemens 8WH1000-0CM07	8WH1000-0CM07	PP1 -X1
6	Terminal 8WH1	Siemens 8WH1000-0AF00	8WH1000-0AF00	PP1 -X1
2	PE terminal 8WH1	Siemens 8WH1000-0CF07	8WH1000-0CF07	PP1 -X1
6	Terminal 8WH1	Siemens 8WH1000-0AK00	8WH1000-0AK00	PP1 -X1
1	PE terminal 8WH1	Siemens 8WH1000-0CK07	8WH1000-0CK07	PP1 -X1

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[WARNING] PP1 / c2: Duty pump (test): Ir 15.2 A, the typical Ir of a 7.5 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate. (motor_current_table; source: policy; engineering confirmation required)
[WARNING] PP1 / c3: Standby pump (test): Ir 56 A, the typical Ir of a 30 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate. (motor_current_table; source: policy; engineering confirmation required)
[WARNING] PP1 / c4: Booster (test): Ir 10.38 A from 5.5 kW shaft power at cos phi 0.85, efficiency 0.9; check it against the nameplate. (motor_current_formula; source: policy; engineering confirmation required)
[WARNING] PP1 / c4: Booster (test): the fuses need at least 75 A prospective short-circuit current at the board to operate fast enough. (drive_fuse_fault_level; source: policy; engineering confirmation required)
[WARNING] PP1: Loads without a power factor were taken at cos phi 0.9. (default_power_factor; source: policy; engineering confirmation required)
[WARNING] PP1: No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze. (earth_fault_no_ze; source: policy; engineering confirmation required)
[WARNING] PP1: Each cable's withstand k^2S^2 is in the cable list; compare it with the selected breaker's let-through I^2t at 10 kA from its maker's curve (ABB handbook §2.4). (short_circuit_withstand; source: policy; engineering confirmation required)
[WARNING] PP1: Circuit rules are this software's defaults, not confirmed by the company's engineers. (rules_unconfirmed; source: policy; engineering confirmation required)
[WARNING] PP1: Enclosure: none named, since 11 devices have no width (rail_widths_mm in the company settings). (enclosure_unknown_widths; source: policy; engineering confirmation required)
[INFO] PP1 / c4: Booster (test): the cable after a drive is protected by the drive's motor overload setting, which is not checked here. (integrity_drive_cable_unchecked; source: policy)
[INFO] PP1 / c1: Panel lighting (test): cable enlarged from 1.5 to 6 mm² for voltage drop over 900 m: 2.81 % from the origin, within the 3 % limit. (voltage_drop_upsized; source: company max_voltage_drop_percent (default IEC 60364-5-52 Annex G, Table G.52.1))
[INFO] PP1 / c1: Panel lighting (test): cable enlarged from 6 to 35 mm² so a short circuit at its far end trips its breaker at once: Ikmin 133 A, at least 100 A (ABB handbook §2.4). (short_circuit_min_upsized; source: ABB Electrical installation handbook Vol. 2, §2.4)
[INFO] PP1: Group 1 breaker raised to 16 A, 1.6 x the 10 A breaker after it, for discrimination. (discrimination_group_raised; source: policy)
[INFO] PP1 / c2: Duty pump (test): dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2. (starter_table; source: policy)
[INFO] PP1 / c3: Standby pump (test): star_delta starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 5: 400 V 50 kA Y/D Normal Type 2. (starter_table; source: policy)
[INFO] PP1 / c3: Standby pump (test): star and delta contactors need a mechanical interlock; the motor is fed by six conductors (U1 V1 W1, U2 V2 W2). (star_delta_interlock; source: policy)
[INFO] PP1 / c4: Booster (test): ACS880-01-12A6-3 for normal duty at 30 °C, behind 25 A aR fuses (ABB ACS880-01 hardware manual (3AUA0000078093)). Use a screened, symmetrical motor cable, earthed 360° at both ends. (drive_selected; source: policy)
[INFO] PP1: Breakers are rated for at least 10 kA breaking capacity against 10 kA prospective at the board; each RCCB's conditional short-circuit current with the breaker ahead of it must reach it too. (breaking_capacity; source: policy)
[INFO] PP1: Leave 1 spare outgoing ways (20 %). (spare_ways; source: policy)
[INFO] PP1: Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables. (discrimination; source: policy)
[INFO] PP1: Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation. (cable_conditions; source: policy)